9/24/2026

LAB NO 03

Operating System

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Class: BSCS F'24 B

LAB TASK

Lab Task 1 — C Program

Solve a C program that accepts a student's name, registration number, and marks in three subjects, **use** appropriate variables and arithmetic operations to calculate the total and percentage, and **show** the student's information, total marks, percentage, and pass/fail status in a properly formatted output. Create, compile, and execute the program using gcc.

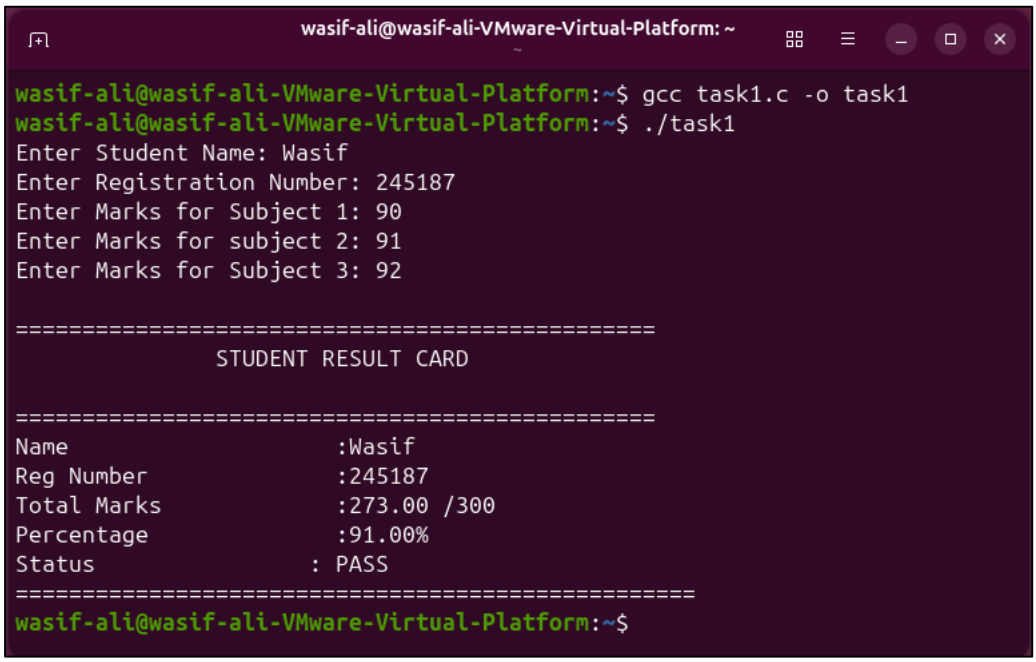
Working:

In this task, we created a C program named task1.c using nano. The program prompts the user to enter the student's name, registration number, and marks for three subjects using scanf. It calculates the total marks and percentage using arithmetic operators. An if-else block checks whether the percentage is 50% or higher to set the pass/fail status. Finally, the program prints a formatted result card using printf. We compiled it using gcc task1.c -o task1 and ran ./task1 to verify the output.

Code and Result:







Lab Task 2 — C++ Program

Complete a C++ program that accepts a student's name, registration number, and marks in three subjects, **use** appropriate variables, arithmetic operations, and an if-else statement to calculate the total and percentage and determine the student's pass/fail status. **Demonstrate** the complete process of creating, compiling, correcting any compilation errors, and executing the program using g++.

Working:

In this task, we developed a C++ program named task2.cpp. We used standard C++ input/output streams (cin and cout) to collect student details and marks. The program evaluates total marks, percentage, and pass/fail condition using an if-else statement. We demonstrated the complete compilation workflow using g++ task2.cpp -o task2 and executed the resulting binary via ./task2. Any compilation syntax errors (such as missing semicolons) were identified and corrected prior to final execution.

Code and Result:

wasif-ali@wasif-ali-VMware-Virtual-Platform: ~ — nano task2.cpp

GNU nano 8.7.1 task2.cpp *

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
	string name, reg_no;
	float m1, m2, m3, total, percentage;

	cout << "Enter Student Name: ";
	getline(cin, name);
	cout << "Enter Registration Number: ";
	cin >> reg_no;
	cout << "Enter Marks for Subject 1: ";
	cin >> m1;
	cout << "Enter Marks for Subject 2: ";
	cin >> m2;
	cout << "Enter Marks for Subject 3: ";
	cin >> m3;

	total = m1 + m2 + m3;
	percentage = (total / 300.0) * 100;

	cout << "\n========================================="<<endl;
	cout << "             STUDENT RESULT CARD             "<<endl;
	cout << "=========================================" <<endl;
	cout << "Name                  : " << name << endl;
	cout << "Reg Number            : " << reg_no << endl;
	cout << "Total Marks           : " << total << " /300" << endl;
	cout << "Percentage            : " << percentage << "%" <<endl;

	if (percentage >= 50.0)
	{
		cout << "Status              : PASS " << endl;
	}
	else
	{
		cout << "Status               : FAIL " << endl;

	}
	cout << "=========================================" << endl;

	return 0;
}
```

wasif-ali@wasif-ali-VMware-Virtual-Platform: ~ — ./task2

```
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$ nano task2.cpp
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$ g++ task2.cpp -o task2
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$ ./task2
Enter Student Name:
```

wasif-ali@wasif-ali-VMware-Virtual-Platform: ~

```
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$ nano task2.cpp
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$ g++ task2.cpp -o task2
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$ ./task2
Enter Student Name: Wasif
Enter Registration Number: 245187
Enter Marks for Subject 1: 99
Enter Marks for Subject 2: 98
Enter Marks for Subject 3: 97

=========================================
             STUDENT RESULT CARD
=========================================
Name                  : Wasif
Reg Number            : 245187
Total Marks           : 294 /300
Percentage            : 98%
Status              : PASS
=========================================
wasif-ali@wasif-ali-VMware-Virtual-Platform:~$
```